

Feature Selection and Rationale

RC Boat Water Quality Project

September 11, 2026

1 Feature Selection Rationale

To ensure the machine learning model operates with maximum efficiency on the embedded STM32 microcontroller, a rigorous feature selection process was conducted. The rationale for including, excluding, or staging each sensor parameter for the final model is detailed below.

1.1 Features Included in the Core Model

The following features were selected because they provide independent, non-redundant informational value to the XGBoost classifier:

- **pH (Hydrogen Ion Concentration):** A foundational indicator of water health. Sudden spikes or drops often indicate industrial chemical dumping or agricultural runoff. Ablation studies proved it provides highly unique, irreplaceable information to the decision trees.
- **Turbidity (Water Cloudiness):** Measures suspended solids (silt, algae, pollutants) using an optical sensor. It is a critical, independent metric for identifying physical pollution and erosion.
- **Electrical Conductivity (EC):** Measures the water's ability to pass an electrical current, which is directly proportional to the concentration of dissolved ions (salts, heavy metals). It is the most heavily weighted feature in the model because it acts as a highly sensitive, broad-spectrum indicator of chemical pollution.
- **Dissolved Oxygen (DO):** The most critical metric for aquatic life. Including DO ensures the model can accurately classify water as "Dangerous" based on immediate biological habitability.
- **Chlorides (CL):** While hardware documentation notes that Chlorides can be roughly approximated via Conductivity, our ablation study empirically proved that the historical chloride data contained enough unique chemical variance to increase the model's overall accuracy by $\approx 0.5\%$. It was retained to maximize predictive precision.
- **Temperature :** Temperature is a vital physical parameter that heavily influences chemical solubilities and biological activity in river ecosystems. It is **not** redundant. However, it was temporarily excluded during this initial model training phase solely

because 98% of the historical dataset lacked temperature readings. Forcing the AI to train on empty data caused statistical noise. In the final deployed model on the RC boat, the live DS18B20 temperature sensor data will be actively collected and permanently added to the feature set to provide crucial context for seasonal water quality variations.

1.2 Features Excluded from the Final Model

The following features were explicitly dropped from the training pipeline to prevent collinearity and data leakage:

- **Total Dissolved Solids (TDS) [Redundant]:** Mathematical and hardware analysis revealed that TDS is a perfect linear derivative of Conductivity ($TDS = EC \times Factor$). Including it caused collinearity (credit hogging), yielding zero new information. Dropping it reduced the computational load on the boat's hardware without sacrificing accuracy.
- **BOD and COD [Data Leakage Hazard]:** Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) are expensive laboratory metrics used exclusively to create the “Answer Key” (y -labels) during data preparation. If the model is allowed to see BOD/COD during training, it causes catastrophic Data Leakage. They are strictly dropped before training to force the AI to learn predictive patterns using only the physical, low-cost sensors.